

Employee Performance & Workforce Analytics

HR Insights across 3,000 Employees | Attrition · Performance · Training · Recruitment

3,000	12.90%	\$1.67M	2.94
TOTAL EMPLOYEES	ATTRITION RATE	TRAINING SPEND	AVG ENGAGEMENT SCORE

DATASET	Synthetic HR Dataset (Kaggle) — 4 relational tables: employees, engagement, training, recruitment
PERIOD	Multi-year employee records across 3,000 employees
RAW DATA	3,000 employee rows per table · joined via EmpID / EmployeeID · ~12,000 total rows across all tables
TOOLSTACK	MySQL 8.0 · MySQL Workbench · Power BI Desktop
TECHNIQUE	Window Functions: RANK, NTILE, CUME_DIST, PERCENT_RANK · CTEs · Z-Score · TIMESTAMPDIFF

Launch Targeted HR Interventions

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1. Executive Summary

3,000 TOTAL EMPLOYEES	2,458 ACTIVE EMPLOYEES	387 TOTAL TERMINATED	12.90% ATTRITION RATE
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387 employees left out of 3,000, which is a 12.90% attrition rate. That number alone does not tell you much. The interesting part is what happens when you break it down. Software Engineering is losing 17.4% of its staff. Production is at 15.9%. Sales is at 3.0%. The same company, three very different attrition stories.

The number that stands out most is the Exceeds performer attrition rate: 10.8%. That is almost identical to Needs Improvement at 11.3%. PIP employees leave at 19.4%, which makes sense. But high performers leaving at roughly the same rate as struggling ones does not. Engagement scores do not explain it either. Exceeds employees average 2.9 on engagement, Fully Meets 3.0, Needs Improvement 2.8, PIP 2.9. The scores are nearly flat across all four tiers. The most likely cause is pay. Zone A employees are the lowest-paid and the flight risk query targets exactly this group: high rating, low pay zone, low engagement.

On training: \$1,675,886 was spent across 3,000 training records at a 50.3% success rate. About \$830K of that produced failed or incomplete outcomes. The recruitment funnel converted 20.33% of applicants to offers, and salary expectations barely varied across education levels — Masters degree holders expected \$65,482 on average, High School graduates \$64,549. A difference of \$933 across four education levels.

Metric	Finding
Total employees	3,000
Active / Terminated	2,458 / 387
Overall attrition rate	12.90%
Highest-attrition department	Software Engineering at 17.4%
Second-highest attrition department	Production at 15.9%
Attrition rate — Exceeds performers	10.8%
Attrition rate — Needs Improvement	11.3%
Attrition rate — PIP employees	19.4%
Avg engagement score (all tiers)	2.8 to 3.0 — flat across performance levels
Total training spend	\$1,675,886
Training success rate	50.3%
Approximate spend on failed outcomes	~\$830,000
Recruitment offer rate	20.33%
Salary spread across education levels	\$933 (Masters \$65,482 to High School \$64,549)

2. Project Setup & Objectives

The Problem

Most HR teams look at each data source separately. Engagement surveys go to one team, training records to another, recruitment to a third. When you join all four tables on EmpID, patterns show up that are invisible in any single table. The clearest example in this dataset: Zone A employees with the highest performance ratings also tend to have the lowest engagement scores. That only appears when you join employees to engagement and filter by pay zone and rating.

What the Analysis Covers

Domain	Business Question	SQL Technique
Performance Ranking	Who are the top-rated employees per department?	RANK(), DENSE_RANK(), ROW_NUMBER()
Performance Baseline	How does each employee compare to their department average?	AVG() OVER (PARTITION BY), deviation
Top Performers	Who sits in the top 10% of ratings company-wide?	PERCENT_RANK() >= 90%
Attrition	Which departments and performance tiers lose the most staff?	Conditional SUM(), GROUP BY
Tenure & Retention	Which long-tenured employees show risk signals?	TIMESTAMPDIFF(), CASE WHEN tiers
Outlier Detection	Who is statistically far above or below their department?	Z-score via STDDEV_POP(), NULLIF
Engagement Correlation	Do high performers report higher engagement scores?	3-table JOIN, AVG() per performance tier
Training ROI	Which departments get the best return on training spend?	Cost-per-success, success rate %
Flight Risk	Who are the high performers most likely to resign?	CTE + Zone A + engagement filter
Recruitment	Does salary expectation affect hiring outcomes?	NTILE(4), PERCENT_RANK(), offer rate

Tables and Relationships

Table	Rows	Key Columns
employees	3,000	EmpID, DepartmentType, PayZone, PerformanceScore, CurrentEmployeeRating, EmployeeStatus, StartDate, GenderCode, EmployeeType
engagement	3,000	EmployeeID (FK to EmpID), EngagementScore, SatisfactionScore, WorkLifeBalanceScore
training	3,000	EmployeeID (FK), TrainingProgramName, TrainingType, TrainingOutcome, TrainingDurationDays, TrainingCost
recruitment	3,000	ApplicantID, EducationLevel, YearsOfExperience, DesiredSalary, JobTitle, Status

3. Dataset & Data Cleaning

Cleaning Steps

Eight cleaning steps were applied before writing any analysis query. Each step has a documented reason. Skipping the documentation makes problems harder to trace later when query results do not match expectations.

Step	Action	Reason
1	Fixed BOM character on EmpID column	The CSV export added a UTF-8 BOM prefix, making the column name unqueryable as written
2	Trimmed trailing spaces in DepartmentType	Spaces caused GROUP BY to return duplicate department rows with separate row counts
3	Converted StartDate from TEXT to DATE	MySQL stores dates from CSV imports as TEXT. TIMESTAMPDIF and date arithmetic need a proper DATE type
4	Converted DOB column with two different date formats	The column held both / and - as separators. A CASE WHEN split handled both formats in one UPDATE
5	Converted ExitDate, preserving NULL for active employees	Active staff have no exit date. A NULL-aware conversion keeps those rows intact
6	Renamed all columns that had spaces in the name	Spaces in column names force backtick quoting on every query. Renaming to camelCase removes that requirement across all four tables
7	Set phone numbers containing # to NULL	Invalid phone numbers in the recruitment table caused downstream issues in string handling
8	Fixed date columns in training and recruitment tables	TrainingDate, ApplicationDate, and DateOfBirth in training and recruitment needed the same STR_TO_DATE treatment as employees
9	Changed column data types with MODIFY COLUMN	Several VARCHAR and DECIMAL columns needed explicit length and type definitions to support indexing and arithmetic correctly

Indexes Added

Ten indexes were created after cleaning. The JOIN key indexes (idx_eng_emp, idx_trn_emp) reduce the cost of every multi-table query. The composite index idx_dept_rating supports the common filter-by-department, order-by-rating pattern without a separate sort step. MySQL requires a length parameter for VARCHAR indexes, so all indexes on text columns include an explicit prefix length.

employees	idx_dept (50), idx_status (40), idx_payzone (10), idx_dept_rating composite (dept+rating), idx_perf (30)
engagement	idx_eng_emp on EmployeeID — primary JOIN key to employees.EmpID
training	idx_trn_emp on EmployeeID (JOIN key), idx_trn_outcome (20)
recruitment	idx_rec_status (30), idx_rec_edu (40)

4. Analysis Methodology

14 analysis queries were written across five domains, plus one stretch goal using NTILE to band employees by performance quartile within each department.

#	Query Name	Domain	Primary Technique
1	Rank employees within department	Performance	RANK(), DENSE_RANK(), ROW_NUMBER()
2	Each employee vs. department average	Performance	AVG() OVER (PARTITION BY), deviation
3	Top 10% performers company-wide	Performance	PERCENT_RANK() filter >= 90%
4	Attrition rate by dept and type	Attrition	Conditional SUM(), GROUP BY
5	Employee tenure analysis	Retention	TIMESTAMPDIFF(), risk tier CASE WHEN
6	Performance score vs. attrition rate	Attrition	Attrition % grouped by performance tier
7	PayZone distribution and attrition	Attrition	Windowed COUNT() for dept %, attrition rate
8	Cumulative rating distribution	Performance	CUME_DIST() per dept and company-wide
9	Z-score outlier detection	Performance	STDDEV_POP(), NULLIF divide-by-zero guard
10	Engagement score vs. performance tier	Engagement	3-table JOIN, AVG() grouped by tier
11	Training ROI — success rate and cost	Training	Cost-per-success, success rate %
12	Zone A high performers — flight risk	Retention	CTE + RANK() + engagement score filter
13	Recruitment funnel and salary banding	Recruitment	NTILE(4), PERCENT_RANK(), offer rate %
14	Executive dashboard query	Summary	Chained CTEs, 3-table LEFT JOIN, median via ROW_NUMBER()
S1	NTILE rating bands and attrition	Stretch	NTILE(4) per dept, attrition rate per quartile

Window Functions Used

Function	How It Was Used
RANK() / DENSE_RANK()	Ranked employees within each department by CurrentEmployeeRating. RANK leaves gaps after ties, DENSE_RANK does not.
ROW_NUMBER()	Used as a workaround for median calculation. MySQL 8.0 has no PERCENTILE_CONT, so ROW_NUMBER assigns positions and the middle row is averaged.
PERCENT_RANK()	Identified the top 10% of performers company-wide by filtering WHERE percentile >= 90.
CUME_DIST()	Shows what proportion of employees score at or below a given value, both within department and company-wide.
NTILE(4)	Splits employees into four equal-size groups by rating within each department, or by salary in the recruitment analysis.
AVG() OVER (PARTITION BY)	Calculates the department average inline so each row can be compared directly to its own department baseline.
STDDEV_POP()	Calculates the population standard deviation per department for the Z-score outlier query.

5. Workforce Results

5.1 Workforce Overview

Production is the biggest department at 2,020 employees, which is 67.3% of the whole workforce. IT/IS is second at 430 and Sales third at 331. The three remaining departments (Software Engineering, Admin Offices, Executive Office) account for 219 employees combined. Employment type is split nearly evenly: Full-Time 1,038, Contract 1,008, Part-Time 954. Female employees make up 56.07% (1,682) and male 43.93% (1,318). Pay zones are also fairly spread: Zone A 1,062 (35.4%), Zone C 985 (32.8%), Zone B 953 (31.77%). On performance, 2,361 employees score Fully Meets, 369 Exceeds, 177 Needs Improvement, and 93 are on PIP.



Figure 1: Workforce Overview — department headcount, employment type, gender, pay zone, and performance score distribution

5.2 Attrition & Performance

387 of 3,000 employees terminated, giving a 12.90% overall attrition rate. Software Engineering is the outlier at 17.4% — with only 115 staff, every departure removes roughly 0.87% of department headcount. Production at 15.9% means around 321 people left. Both are above the company average. Sales and Admin Offices are well below it at 3.0% and 1.3%.

Looking at attrition by performance score tells a different story. PIP employees leave at 19.4% as expected. What is harder to explain is that Exceeds employees leave at 10.8%, which is close to Needs Improvement at 11.3%. The engagement scores across all four tiers range from only 2.8 to 3.0, so performance tier is not driving engagement in a meaningful way. Male employees leave at a slightly higher rate (13.91%) than female (11.61%). Zone A and Zone B both sit at 14% attrition, Zone C at 11%.

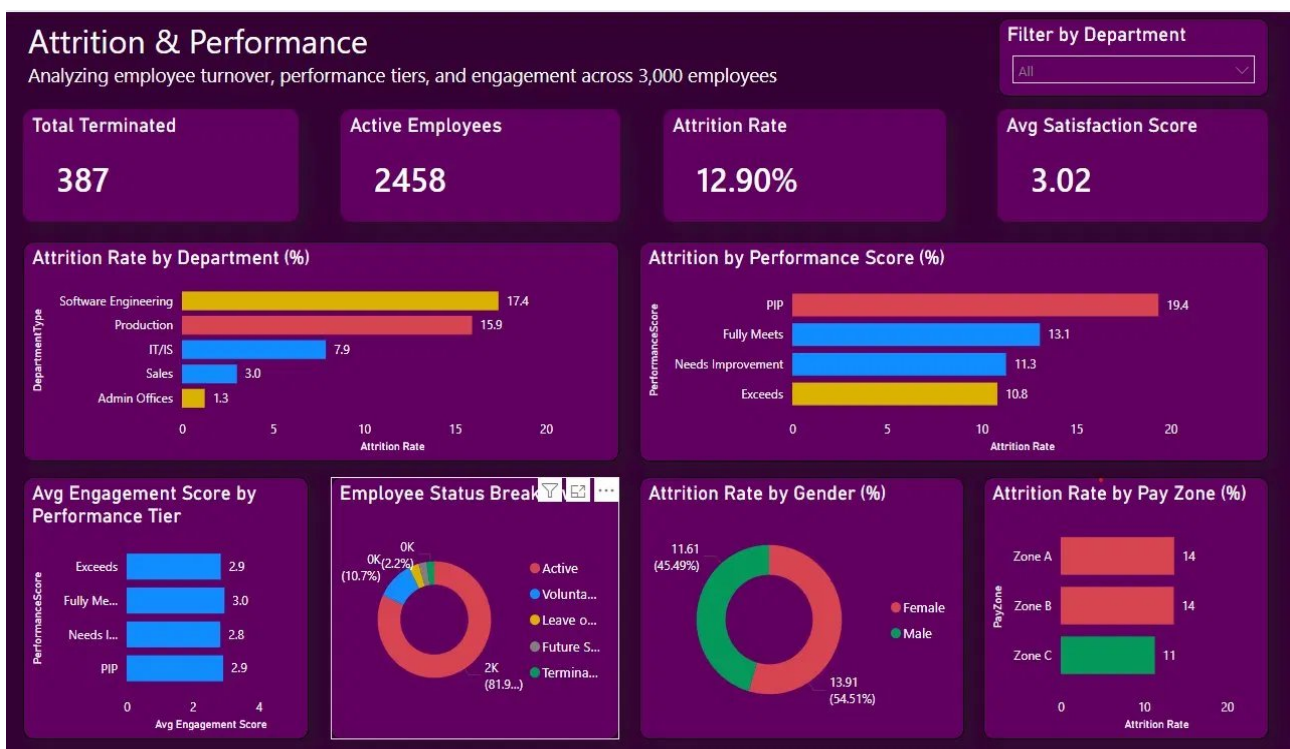


Figure 2: Attrition & Performance — attrition by department, performance score, gender, pay zone, and engagement score by tier

5.3 Training & Recruitment

Total training spend came to \$1,675,886 across all 3,000 training records. Production accounts for \$1.13M of that, which reflects its size. Overall training success rate is 50.3%. Admin Offices has the best success rate at 53% despite the smallest training budget (~\$40K). Training is split 50/50 between Internal (1,491 sessions) and External (1,509 sessions).

The recruitment funnel is unusually balanced. Applied (611), Offered (610), In Review (595), Rejected (594), and Interviewing (590) are all within 21 of each other. Overall offer rate is 20.33%, or roughly 1 in 5 applicants. Desired salary across education levels is almost identical: Masters degree \$65,482, High School \$64,549. A \$933 gap across four education levels is not a meaningful difference.

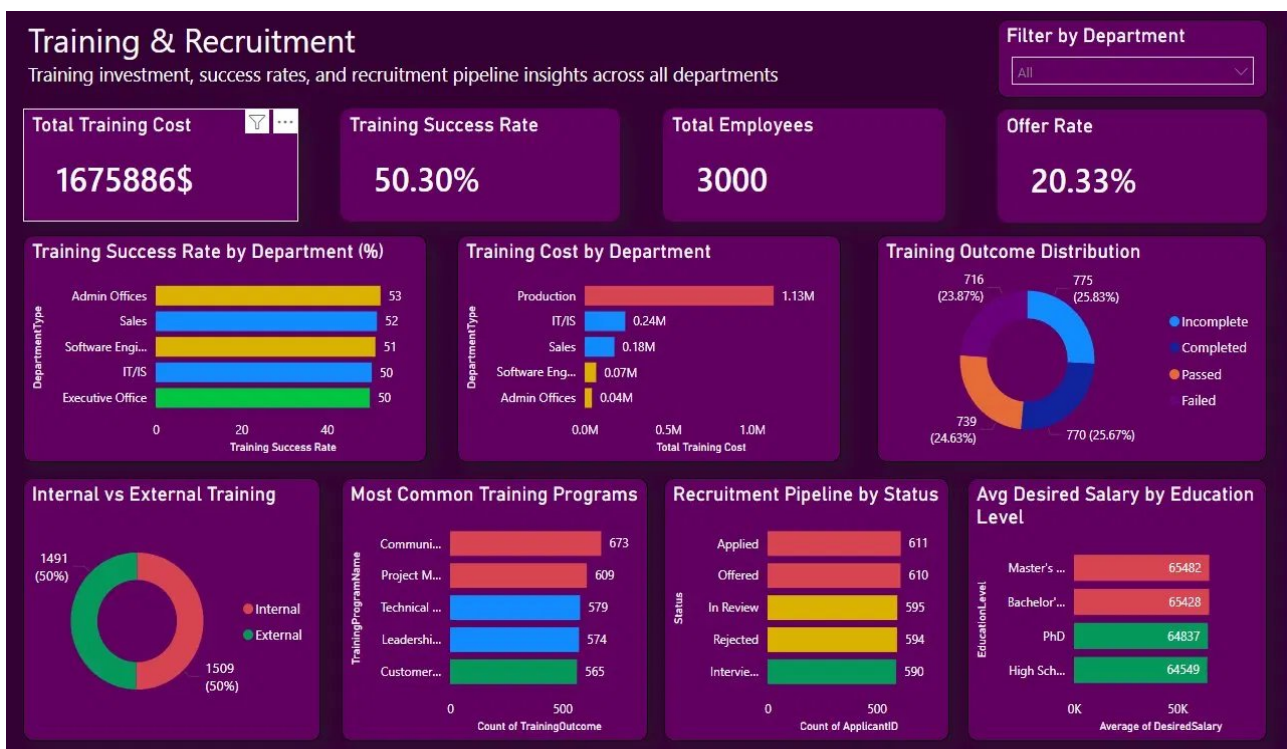


Figure 3: Training & Recruitment — training success rates, cost by department, outcome distribution, recruitment pipeline, salary by education level

6. Business Questions

Q1 — Which departments have the highest attrition?

17.4% SOFTWARE ENG	15.9% PRODUCTION	7.9% IT/IS	3.0% SALES
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Software Engineering has the highest rate at 17.4%. It is also the fourth-smallest department with 115 employees, so each departure has an outsized impact. Production at 15.9% is a volume problem: 321 people leaving from a 2,020-person department is a significant operational strain even if the percentage looks manageable. IT/IS at 7.9% is below the company average. Sales at 3.0% and Admin Offices at 1.3% are the only departments that look genuinely healthy on this metric.

Department	Headcount	Attrition Rate	Note
Software Engineering	115	17.4%	Each departure removes 0.87% of department capacity
Production	2,020	15.9%	Roughly 321 people left — largest absolute volume
IT/IS	430	7.9%	Below company average, not an immediate concern
Sales	331	3.0%	Second-largest department, second-lowest attrition
Admin Offices	80	1.3%	Lowest attrition rate in the company
Executive Office	24	N/A	Sample too small for a reliable rate

Note: Software Engineering and Production together account for 2,135 employees, which is 71.2% of the workforce. Both sit above the 12.9% company average. Addressing those two departments covers the majority of the attrition problem by volume.

Q2 — Does performance tier predict who leaves?

Partially. PIP employees leave at 19.4%, the highest rate, which makes sense. But Exceeds employees leave at 10.8%, which is almost the same as Needs Improvement at 11.3%. If performance issues were the main driver, you would expect a much larger gap between those two tiers. The fact that they are within 0.5 percentage points of each other suggests something else is at play, most likely compensation, since pay zone attrition tells a similar story: Zone A and Zone B both sit at 14%, Zone C at 11%.

Performance Score	Total Employees	Attrition Rate	Interpretation
PIP	93	19.4%	Highest rate — expected for employees on formal plans
Fully Meets	2,361	13.1%	Largest group; most departures come from here by volume
Needs Improvement	177	11.3%	Moderate — some natural exit for underperformers
Exceeds	369	10.8%	Nearly identical to Needs Improvement — the key finding

Key finding: Exceeds performers leaving at 10.8% is not a performance problem. It is a pay or engagement problem. These employees have high output and a demonstrated track record. They are also, in many cases, sitting in Zone A. The flight risk query in Q3 targets this exact population.

Q3 — Who are the highest-priority flight risk employees?

Query 12 isolates active employees who have three things in common: they are in Zone A (lowest pay tier), their CurrentEmployeeRating is 4 or above, and their PerformanceScore is Exceeds or Fully Meets. Those employees are then grouped by engagement and satisfaction score into three action tiers:

Label	Score Criteria	Suggested Action
CRITICAL — Actively Disengaged	EngagementScore <= 2 AND SatisfactionScore <= 2	Compensation review and 1-on-1 within two weeks. Highest exit probability.
HIGH RISK — Low Engagement	EngagementScore <= 3	Manager conversation within 30 days. Retention discussion if budget allows.
MONITOR	EngagementScore > 3	Quarterly check-in. Track for score decline in next survey cycle.

Zone A employees are the ones with the most reason to leave. They are doing high-rated work at the lowest pay tier. In Software Engineering, where attrition is already 17.4%, losing a Zone A Exceeds performer is a particularly costly exit because that type of employee is both hard to replace and probably getting offers elsewhere.

Q4 — Is the \$1.67M training budget producing results?

\$1.67M TOTAL SPEND	50.3% SUCCESS RATE	~\$830K ON FAILURES	53% BEST DEPT RATE
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At 50.3%, training is almost exactly split between successful and unsuccessful outcomes. Production absorbed \$1.13M, which is 67.4% of the entire training budget. Admin Offices had the best success rate at 53% on roughly \$40K of spend. The raw success rate by department does not vary much: the range is 50% to 53% across all departments. The real question is which specific training programs are dragging the rate down, and that requires a cost-per-success breakdown by program name rather than by department.

Department	Training Cost	Success Rate	Note
Production	~\$1,130,000	~50%	67.4% of the total training budget
IT/IS	~\$240,000	50%	Average success rate
Sales	~\$180,000	52%	Above average on mid-range spend
Software Engineering	~\$70,000	51%	Relatively small spend, average success
Admin Offices	~\$40,000	53%	Best success rate, lowest departmental cost

Recommendation: Around \$830K was spent on training sessions that ended with Failed or Incomplete outcomes. Query 11 calculates cost_per_success by department, but the more useful cut would be by TrainingProgramName. That would show which programs should be cut and which should receive more budget.

Q5 — What does the recruitment pipeline show?

The five pipeline statuses (Applied, Offered, In Review, Rejected, Interviewing) are spread within a 21-applicant range of each other. Applied 611, Offered 610, In Review 595, Rejected 594, Interviewing 590. The offer rate across the whole pipeline is 20.33%.

Part A — Average desired salary by education level

Education Level	Avg Desired Salary	Note
Master's Degree	\$65,482	Highest average expectation
Bachelor's Degree	\$65,428	\$54 below Masters — functionally the same
PhD	\$64,837	Lower than Masters and Bachelor's combined
High School	\$64,549	Only \$288 below PhD

Finding: The \$933 spread across all four education levels is small enough to be noise. PhD applicants expect less than Masters and Bachelors applicants. Education level is not functioning as a meaningful salary signal in this applicant pool, which raises a question about how job descriptions are written and whether salary bands are differentiated by qualification.

Part B — NTILE(4) salary quartile banding

Applicants are split into four equal groups using NTILE(4) on DesiredSalary. The overall 20.33% offer rate holds consistently across all four bands. Salary expectation level does not materially affect hiring outcomes in this dataset.

Salary Quartile	Approx. Range	Approx. Applicants	Note
Q1 — Bottom 25%	\$30K to \$47K	~750	Lowest salary expectations, no offer rate advantage
Q2	\$47K to \$65K	~750	Lower-mid band
Q3	\$65K to \$82K	~750	Upper-mid band
Q4 — Top 25%	\$82K to \$100K	~750	Highest expectations, offer rate similar to Q1

7. HR Strategy Recommendations

Each finding maps to a specific action. The table below is ordered by how urgent the business impact is, not by how easy the action is to implement.

Priority	Finding	Recommended Action	Timeline
Critical	Zone A, high-rated, low-engagement employees	Run the flight risk query. Anyone labelled CRITICAL gets a compensation review and a direct manager conversation within two weeks.	14 days
Urgent	Software Engineering attrition at 17.4%	Exit interview audit for the department. Identify whether leavers are voluntary or for cause and what roles are hardest to backfill.	30 days
Urgent	Exceeds performers leaving at 10.8%	Structured stay interviews for all current Exceeds-rated employees in Zone A. Ask directly about pay and intent to stay.	Ongoing
High	Training: \$830K on failed outcomes	Break down cost_per_success by TrainingProgramName, not just by department. Cut lowest-ROI programs and reallocate budget.	60 days
High	Z-score outliers with $z > +2$	Fast-track promotion or pay review for employees statistically far above their department average.	60 days
High	Z-score outliers with $z < -2$	Structured improvement plan with defined criteria and a clear timeline for review.	90 days
Medium	Avg engagement 2.94 across all tiers	Engagement scores are flat across performance tiers (2.8 to 3.0). This needs a company-wide response, not a targeted department fix.	Q2
Low	Salary spread of \$933 across education levels	Review job descriptions. Consider whether qualification level should carry more weight in salary banding.	Quarterly

Note on engagement: Engagement scores for Exceeds employees average 2.9. For PIP employees they average 2.9 as well. The same number, opposite ends of the performance spectrum. This tells you the engagement problem is not driven by performance tier, it is structural. Fixing it through performance management alone will not work.

8. Conclusion

387 people left out of 3,000 at a 12.90% rate. The company average masks what is happening in specific departments. Software Engineering at 17.4% and Production at 15.9% are both above average. Software Engineering is the more urgent problem per departure given its small headcount. Production is the bigger operational concern by volume.

The Exceeds performer attrition rate of 10.8% is the finding that needs the most attention. High performers leaving at nearly the same rate as Needs Improvement employees (11.3%) is not explained by engagement scores since those are flat across all tiers. The flight risk query specifically targets this group: Zone A pay, high performance rating, low engagement. Acting on that list before those employees decide to leave is where this analysis has the most direct return.

\$1.67M in training spend at 50.3% success means around \$830K went to outcomes labelled Failed or Incomplete. The department-level breakdown in Query 11 is a start, but the program-level cut is what would actually tell you which training programs to keep and which to cut.

Metric	Result
Employees analysed	3,000
Active / Terminated	2,458 / 387
Overall attrition rate	12.90%
Highest-attrition department	Software Engineering at 17.4%
Production attrition	15.9% (~321 people)
Exceeds performers attrition rate	10.8%
Needs Improvement attrition rate	11.3% (0.5pp gap from Exceeds)
PIP attrition rate	19.4%
Avg engagement — Exceeds performers	2.9 / 5
Avg engagement — PIP employees	2.9 / 5 (same score, opposite tiers)
Total training spend	\$1,675,886
Training success rate	50.3%
Approx. spend on failed outcomes	~\$830,000
Recruitment offer rate	20.33%
Salary spread by education level	\$933 (Masters \$65,482 to High School \$64,549)
Recommended immediate action	Run flight risk query. Contact CRITICAL tier employees within 14 days.